Table of contents

Series 8: Introduction to Quantifiers and Transformations in First-Order Logic

Methods

Satisfiability and Validity Verification in First-Order Logic

- **Satisfiability**: Find an interpretation (domain, functions, predicates) that makes the formula true.
- **Validity**: Verify if the formula is true for all interpretations.
- **Counterexample**: To show a formula is not valid, exhibit an interpretation that makes it false.

Elimination of Connectives and Transformation to Normal Form

- **Elimination of equivalence ($\iff$)**: Replace $A \iff B$ with $(A \Rightarrow B) \land (B \Rightarrow A)$.
- **Elimination of implication ($\Rightarrow$)**: Replace $A \Rightarrow B$ with $\neg A \lor B$.
- **Prenex form**: Move all quantifiers to the front of the formula.
- **Skolemization**: Replace existential variables with Skolem functions (or constants) to eliminate existential quantifiers and obtain a universal form (clauses).

Inductive Definition of Free Variables (FV)

- **Free variables**: Variables not quantified in a formula.
- **Inductive definition**: By recursion on the structure of terms and formulas.

Substitutions and Closed Instances

- **Substitution**: Replacement of variables by terms.
- **Closed instance**: Result of a substitution that eliminates all free variables (yields a formula without free variables).

Validity of Clauses Under Substitution

- **Valid clause**: A clause is valid if it is satisfied for every interpretation in a given domain with fixed functions and predicates.
- **Effect of substitution**: If a clause is valid, then any substitution (in particular renaming) applied to that clause preserves validity.

Reminder

- **Implication law**: $A \Rightarrow B \equiv \neg A \vee B$.
- **Equivalence law**: $A \iff B \equiv (A \Rightarrow B) \wedge (B \Rightarrow A)$.
- **Quantifier distributivity**:
  - $\forall x(P(x) \vee Q) \equiv (\forall x P(x)) \vee Q$ if x is not free in Q.
  - $\exists x(P(x) \wedge Q) \equiv (\exists x P(x)) \wedge Q$ if x is not free in Q.
- **Negation equivalences**:
  - $\neg \forall x P(x) \equiv \exists x \neg P(x)$.
  - $\neg \exists x P(x) \equiv \forall x \neg P(x)$.
- **Skolemization**: For $\exists y \forall x Q(x, y)$, introduce a Skolem function f such that $y = f(x)$ (if the existential quantifier is after the universal ones). If the existential quantifier is at the front, introduce a constant.
- **Universal form**: A formula of the form $\forall x_1 \forall x_2 ... \forall x_n C$ where C is a clause (disjunction of literals).
- **Validity of a clause**: A clause C is valid if for every interpretation $\mathcal{I}$, $\mathcal{I} \vDash C$.

Illustrative Examples

Example 1: Satisfiability and Validity Verification

Context: Consider the formula $\forall x p(x) \iff \neg p(f(x))$.

Application:

1. **Satisfiability**: Choose the following interpretation:

   - Domain $\mathcal{D} = \mathbb{N}$.
   - $f(x) = x + 1$.
   - $p(x)$: "x is even".

   Then:

   - If x is even, $p(x)$ is true, $f(x) = x + 1$ is odd, so $\neg p(f(x))$ is true. Thus, $p(x) \iff \neg p(f(x))$ is true (T $\iff$ T).
   - If x is odd, $p(x)$ is false, $f(x) = x + 1$ is even, so $\neg p(f(x))$ is false. Thus, $p(x) \iff \neg p(f(x))$ is true (F $\iff$ F).

   Therefore the formula is satisfied for all x, hence satisfiable.

2. **Validity**: To show it is not valid, take a different interpretation:

   - Domain $\mathcal{D} = \{0\}$.
   - $f(0) = 0$.
   - $p(0)$ true. Then $p(0)$ true and $\neg p(f(0)) = \neg p(0)$ false, so the equivalence is false. The formula is therefore not valid.

Example 2: Transformation to Universal Form

Context: Transform the formula $\exists x p(x) \iff \forall y \exists z q(y, z)$ into universal form.

Application:

1. Eliminate $\iff$:

$$(\exists x p(x) \Rightarrow \forall y \exists z q(y, z)) \wedge (\forall y \exists z q(y, z) \Rightarrow \exists x p(x))$$

2. Eliminate $\Rightarrow$ (using $A \Rightarrow B \equiv \neg A \vee B$):

$$(\neg \exists x p(x) \vee \forall y \exists z q(y, z)) \wedge (\neg \forall y \exists z q(y, z) \vee \exists x p(x))$$

3. Move negations inside:

$$(\forall x \neg p(x) \vee \forall y \exists z q(y,z)) \wedge (\exists y \forall z \neg q(y,z) \vee \exists x p(x))$$

4. Rename variables to avoid conflicts:

$$(\forall x_1 \neg p(x_1) \vee \forall y_1 \exists z_1 q(y_1,z_1)) \wedge (\exists y_2 \forall z_2 \neg q(y_2,z_2) \vee \exists x_2 p(x_2))$$

5. Skolemization:

   - For $\exists z_1$ in the first disjunction, we have $\forall y_1 \exists z_1 q(y_1, z_1)$. Introduce a Skolem function f such that $z_1 = f(y_1)$. We obtain $\forall y_1 q(y_1, f(y_1))$.
   - For $\exists y_2$ and $\exists x_2$ in the second disjunction, introduce constants a and b: $y_2 = a$ and $x_2 = b$. We then have:

$$(\forall x_1 \neg p(x_1) \vee \forall y_1 q(y_1, f(y_1))) \wedge (\forall z_2 \neg q(a, z_2) \vee p(b))$$

6. Put into prenex form (all quantifiers at the front):

$$\forall x_1 \forall y_1 \forall z_2 [(\neg p(x_1) \vee q(y_1, f(y_1))) \wedge (\neg q(a, z_2) \vee p(b))]$$

   This formula is now in universal form (conjunction of universally quantified clauses).

Example 3: Inductive Definition of Free Variables (FV)

Context: Give an inductive definition of $\mathrm{FV}(\phi)$ for a first-order formula ϕ.

Application: First define FV for terms:

- $\mathrm{FV}(x) = \{x\}$ for a variable x.
- $\mathrm{FV}(c) = \emptyset$ for a constant c.
- $\mathrm{FV}(f(t_1, ..., t_n)) = \bigcup_{i=1}^{n} \mathrm{FV}(t_i)$.

Then for formulas:

- $\mathrm{FV}(p(t_1, ..., t_n)) = \bigcup_{i=1}^{n} \mathrm{FV}(t_i)$.
- $\mathrm{FV}(\neg\phi) = \mathrm{FV}(\phi)$.
- $\mathrm{FV}(\phi \wedge \psi) = \mathrm{FV}(\phi) \cup \mathrm{FV}(\psi)$ (similarly for $\vee, \Rightarrow, \iff$).
- $\mathrm{FV}(\forall x \phi) = \mathrm{FV}(\phi) \setminus \{x\}$.
- $\mathrm{FV}(\exists x \phi) = \mathrm{FV}(\phi) \setminus \{x\}$.

Example: For $\phi = \forall x p(x) \wedge q(y)$, we have:

- $\mathrm{FV}(p(x)) = \{x\}$, so $\mathrm{FV}(\forall x p(x)) = \emptyset$.
- $\mathrm{FV}(q(y)) = \{y\}$.
- Therefore $\mathrm{FV}(\phi) = \emptyset \cup \{y\} = \{y\}$.

Example 4: Substitutions to Obtain Closed Instances

Context: Let $\phi = p(x) \iff \neg p(f(x))$. Propose three substitutions θ such that $\phi.\theta$ is a closed instance of ϕ.

Application: A closed instance has no free variables. Here, x is free in ϕ (because it is not quantified in ϕ itself, though in the exercise we have $\forall x \phi$, but here we consider ϕ alone). To close ϕ, we must substitute x with a term without variables.

1. $\theta_1 = \{x \mapsto a\}$ where a is a constant. Then $\phi.\theta_1 = p(a) \iff \neg p(f(a))$.
2. $\theta_2 = \{x \mapsto f(a)\}$. Then $\phi.\theta_2 = p(f(a)) \iff \neg p(f(f(a)))$.
3. $\theta_3 = \{x \mapsto f(f(a))\}$. Then $\phi.\theta_3 = p(f(f(a))) \iff \neg p(f(f(f(a))))$.

Each of these instances is closed because it contains no more variables.

Example 5: Validity of Clauses Under Substitution

Context: Let C be a valid clause in a context $(\mathcal{D}, \mathcal{F}, \mathcal{P})$. Let Π be a substitution on the variables of C. Prove that if C is valid, then $\Pi(C)$ is valid. If Π is a renaming (bijection between variables), prove the converse.

Application:

1. **Direct proof**:

   If C is valid, then for every interpretation $\mathcal{I}$, $\mathcal{I} \models C$.

   Let $\mathcal{I}$ be an interpretation. We want to show $\mathcal{I} \models \Pi(C)$.

   Let σ be a variable assignment. Define an assignment σ' such that for every variable x, $\sigma'(x) = \mathcal{I}(\Pi(x))$ (the value of $\Pi(x)$ in $\mathcal{I}$).

   Then $\mathcal{I}, \sigma \models \Pi(C)$ if and only if $\mathcal{I}, \sigma' \models C$.

   Since C is valid, $\mathcal{I}, \sigma' \models C$, therefore $\mathcal{I}, \sigma \models \Pi(C)$. Thus $\Pi(C)$ is valid.

2. **Converse for renaming**:

   If Π is a renaming (bijection), then Π^{-1} exists and is also a substitution.

   If $\Pi(C)$ is valid, then applying the previous result with Π^{-1}, we obtain that $\Pi^{-1}(\Pi(C)) = C$ is valid.

Concrete example: Let $C = P(x) \vee \neg P(f(x))$. Assume C is valid. Consider substitution $\Pi = \{x \mapsto f(y)\}$. Then $\Pi(C) = P(f(y)) \vee \neg P(f(f(y)))$. According to the proof, $\Pi(C)$ is also valid.

Synthesis

The exercises in this series implement basic techniques of first-order logic:

- Distinction between satisfiability and validity, with construction of interpretations and counterexamples.
- Formula transformations: elimination of connectives, conversion to prenex form, skolemization to obtain universal forms (clauses).
- Manipulation of free variables and substitutions to obtain closed instances.
- Preservation of clause validity under substitution, particularly under renaming.

These methods are fundamental for resolution in first-order logic, automated theorem proving, and software verification.